



Physics. *You work it out.*

Waves and Complex numbers

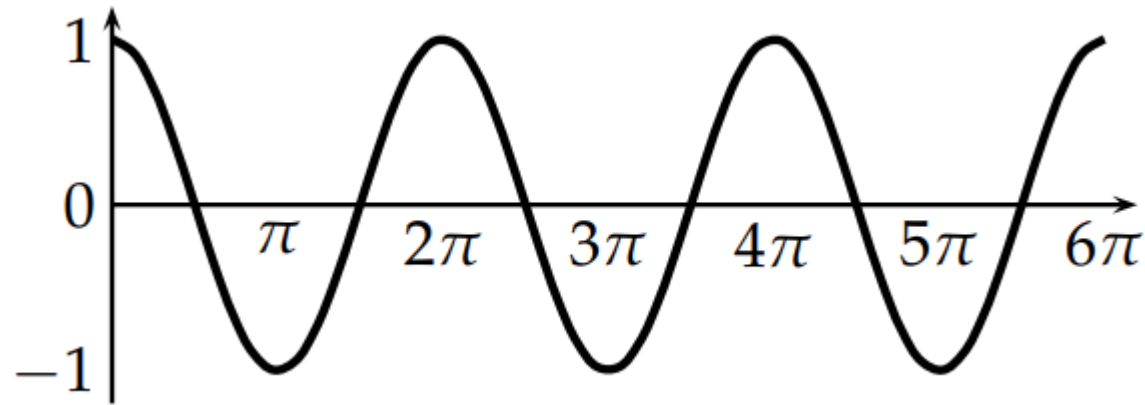
Overture for Senior Physics Challenge

isaacphysics.org



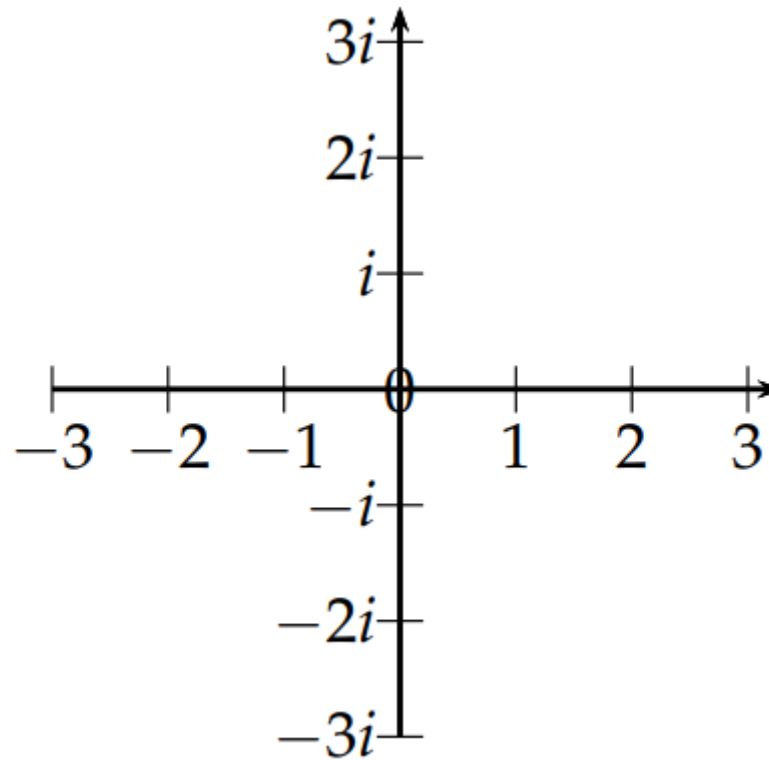
Describing a wave

- › Phase gives 'location' within wave 0=peak.
- › Angles are always measured in radians ($1 \text{ rad} = 180/\pi^\circ$)

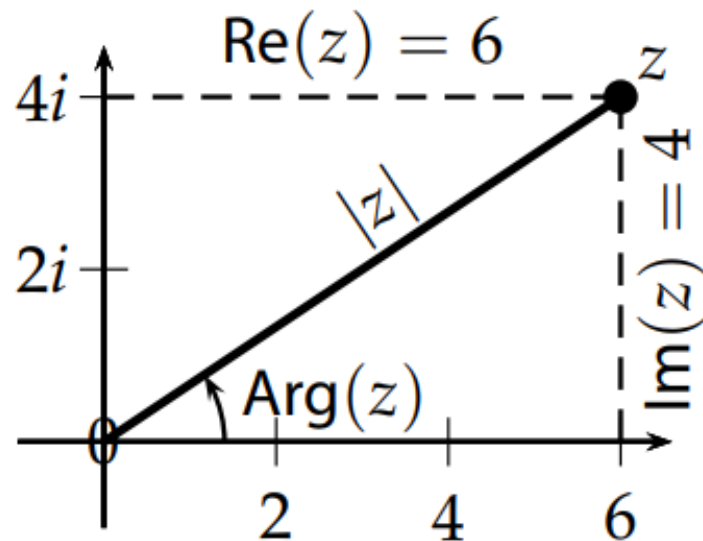


- › We describe the wave $y = A \cos \omega t$
- › Angular frequency $\omega = 2\pi f = 2\pi/T$ measured in rad/s

The Argand diagram



A complex number



This number is $z = 6 + 4i$

Two complex numbers are added with usual algebra

$$3 + 2i + 6 - 3i = 9 - i$$

Multiplication by i represents rotation by $\frac{\pi}{2}$ anticlockwise.

$$i^2 = -1, \quad -i \times i = 1, \quad \frac{1}{i} = -i$$

Complex conjugate $z^* = 6 - 4i$, $z^* z = |z|^2$

Euler: $z = r \cos \theta + ir \sin \theta = r e^{i\theta}$, so $re^{i\theta} we^{i\phi} = rwe^{i(\theta+\phi)}$



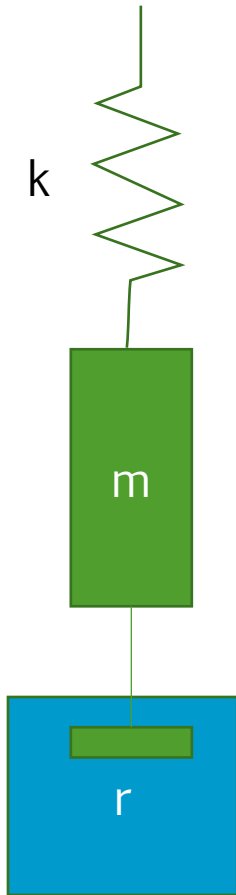
Using complex numbers

- › Using $e^{i\theta} = \cos \theta + i \sin \theta$, work out
- › $e^{i\theta} + e^{-i\theta}$
- › $e^{i\theta} - e^{-i\theta}$
- › $\sin(a+b)$
- › $\sin a + \sin b$
- › Find the modulus of $ae^{i\phi} + be^{i\theta}$

Forced, damped harmonic oscillator

$$F = m\ddot{x} = -kx - r\dot{x} + F_0 \cos \omega t$$

Try solution $z = Ae^{i\omega t}$ to $m\ddot{z} = -kz - r\dot{z} + F_0e^{i\omega t}$

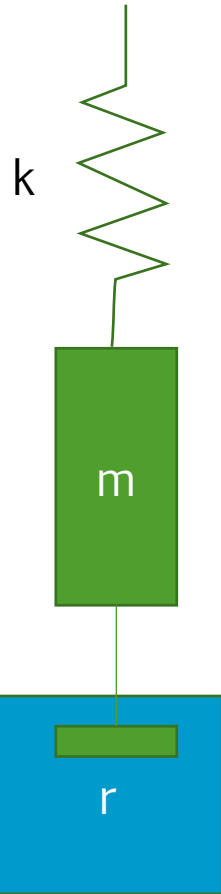


$$\frac{d}{dt}e^{i\omega t} = i\omega e^{i\omega t} \quad \frac{d^2}{dt^2}e^{i\omega t} = -\omega^2 e^{i\omega t}$$

Forced, damped harmonic oscillator

$$F = m\ddot{x} = -kx - r\dot{x} + F_0 \cos \omega t$$

Try solution $z = Ae^{i\omega t}$ to $m\ddot{z} = -kz - r\dot{z} + F_0e^{i\omega t}$



$$\frac{d}{dt}e^{i\omega t} = i\omega e^{i\omega t} \quad \frac{d^2}{dt^2}e^{i\omega t} = -\omega^2 e^{i\omega t}$$

$$-m\omega^2 Ae^{i\omega t} = -kAe^{i\omega t} - ri\omega Ae^{i\omega t} + F_0e^{i\omega t}$$

$$(k - m\omega^2 + ir\omega)A = F_0$$